

North America, while the western mountains will certainly harbour a much greater number of unknown forms.

The group of metallic blue species of *Coelichneumon* deserve special attention. In the Nearctic there are a surprising number of different forms which in their similarity of color represent a taxonomic puzzle. Much more material and research will be needed in order to recognize the limits of individual mutability and regional variability of these blue species.

In the following keys the term "white" will be used in a generalized sense covering all shades between pure white and yellow.

### 1. Genus *Protichneumon* Thomson

*Protichneumon* Thomson, 1893, Opusc. Ent., fasc. 18, p. 1899.

*Type-species.*—*Ichneumon fusorius* Linnaeus, designated by Ashmead, 1900.

The genus shares all general characters with *Coelichneumon* Thomson, in particular the strongly oxypygous abdomen of the female, the pronounced, deep gastrocoeli, the striate or striate-punctate sculpture of the postpetiolus. It is morphologically distinguished from *Coelichneumon* by nothing else than a slight deviation of the propodeal areolation: the area superomedia—usually horseshoe-shaped or with the outline of a gothic arch—is raised anteriorly a little above the level of the horizontal part of the propodeum, so that the areae superoexternae slope very slightly from the borders of the anterior part of the area superomedia toward the sides and simultaneously toward the base of the propodeum. This character however occurs also in the Oriental genus *Aglaojoppa* Cameron, which is hard to distinguish from *Coelichneumon*. The most remarkable character of the *Protichneumon* species is their considerable size. In this particular case size is not a mere casual feature but the manifestation of an important biological character: the specialization of this genus for Sphingidae as its host. The host specialization and the size, as indicative of the latter, seems to be the only character justifying the separation of *Protichneumon* and *Coelichneumon*.

### Key to the Species of the Genus *Protichneumon* Thomson of America North of Mexico

#### Females and Males

1. Legs III entirely black. .... 2  
    Legs III entirely or at least the tibiae III red or reddish brown. .... 4
2. Scopa very large, trimmed flat on top; postpetiolus always red; scutellum more raised and more densely punctured than in the alternative species; widest segment of female flagellum hardly more than twice as wide as long; tyloides of male flagellum narrower than in the alternative species; size 17-20 mm. (Scutellum and pronotal ridge not white marked.) ..... 2. *effigies*, new species  
    Scopa small, not very dense, exceptionally absent; postpetiolus black, exceptionally obscure reddish; scutellum less raised, sparsely punctured and shiny; widest segment of female flagellum three times as wide as long; tyloides of male flagellum wider than in the alternative species; size 19-25 mm. (Scutellum and pronotal ridge often white marked.) ..... 3
3. Tergites 2-4 normally chitinized, not strongly separated nor bulging laterally, with normally dense, not very strong sculpture. (Scutellum and pronotal ridge often white marked.) ..... 1a. *grandis grandis* Brullé  
    Tergites 2-4 very strongly chitinized, sharply separated, sometimes slightly bulging laterally toward the apex, with extremely dense and gross sculpture. (Scutellum and pronotal ridge seldom white marked.) ..... 1b. *grandis regnatrix* Cresson
4. Apical half of postpetiolus, tibiae and tarsi III and the apical part of femora III reddish-brown; tyloides of male flagellum relatively wide-oval, the longest reaching almost from base to apex of the segments. (Scutellum of the male white marked; length 23-26 mm.) ..... 1c. *grandis victoriae*, new subspecies  
    Entire first segment and entire femora and tibiae III bright red, tarsi III dark; tyloides of male flagellum narrower and shorter than in the alternative species. (Scutellum not white marked in the male; length 19 mm.) ..... 3. *polytropos*, new species